Konekt Telcom

USE CASE DOCUMENT

Incident Response · Business Continuity · Disaster Recovery

BC/DR/IR Program— Konekt Telcom

⚠ SCENARIO: OPERATION BLACKOUT

A coordinated ransomware attack cripples Konekt Telcom's primary data centre, triggering a simultaneous P1 Network Outage and P1 Cybersecurity Incident — affecting 2.3 million subscribers.

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Policy Owner	Samara Risha, Aya Reda , Abdallah Khaled	**CONFIDENTIAL**
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EXECUTIVE SUMMARY

This document presents a comprehensive use case for Konekt Telcom's Business Continuity, Disaster Recovery, and Incident Response (BC/DR/IR) program It narrates a realistic, high-stakes scenario — Operation Blackout — that stress-tests every pillar of the program simultaneously, and demonstrates exactly how Konekt Telcom's 15-document framework enables the organization to detect, contain, recover, and emerge stronger.

Why This Use Case Matters

In telecommunications, a 1-hour P1 outage costs Konekt Telcom EGP 200,000+ in direct revenue loss, exposes the company to TRA regulatory fines, and risks permanent subscriber churn. This use case proves that the BC/DR/IR program is not a binder on a shelf — it is a living, tested, operational capability that has been designed to survive the worst.

PROGRAMME DOCUMENT MAP

The following 15 documents constitute the complete Konekt Telcom BC/DR/IR program. Each is referenced throughout this use case at the moment it is activated:

Doc No.	Document Name	Purpose	Activated In
00-KNKT-OVW	Company Overview	Organizational context, services, RTO/RPO objectives	Background
01-KNKT-SCOPE	ISMS Scope	Defines what is in-scope; all locations, depts, personnel	Background
02-KNKT-STKS	Stakeholder Register	Who gets notified, when, and what they need	Phase 2–4
04-KNKT-ASSET	Asset Inventory	Identifies every critical system and its owner	Phase 1–2
05-KNKT-RISK	Risk Assessment	Pre-identified ransomware & outage risk scenarios	Background
06-KNKT-BIA	Business Impact Analysis	RTO, RPO, MTD, financial impact per business function	Phase 1–3
07-KNKT-CCP	Crisis Communication Plan	Stakeholder messaging, templates, channels, timelines	Phase 2–5
08-KNKT-CONT	Emergency Contact Tree	Who to call, in what order, with backup contacts	Phase 1–2
09-KNKT-BCP-NET	BCP – Core Network	Step-by-step network recovery, DR failover procedures	Phase 2–3
10-KNKT-BCP-CIR	BCP – Cyber IR	Ransomware, DDoS, breach response playbooks	Phase 1–4
11-KNKT-BCP-BILL	BCP – Billing	Billing continuity, CDR recovery, Amdocs failover	Phase 3
12-KNKT-DRTEST	DR Test Plan	Testing schedule, scenarios, results register	Phase 5
13-KNKT-PIR	Post-Incident Review	Lessons learned template, RCA, corrective actions	Phase 5
14-KNKT-SOA	Statement of Applicability	93 ISO 27001 controls — status, justification, evidence	Audit
15-KNKT-TRAIN	Training Plan	Staff competence — the human layer of resilience	Pre-incident

THE SCENARIO — OPERATION BLACKOUT

Setting the Scene

It is 02:47 on a Tuesday morning. Cairo is quiet. Konekt Telcom's Network Operations Centre (NOC) in Nasr City hums with the usual overnight rhythm — two analysts monitoring dashboards, one engineer on a maintenance window.

At exactly 02:47:13, IBM Radar SIEM (Asset SV-005) fires 47 simultaneous alerts. Within seconds, the NOC monitors go dark. The primary Cisco ASR 9000 core router (NI-001) stops responding. BGP sessions to all three upstream ISP peers drop simultaneously. Subscriber connectivity flatlines across Cairo, Alexandria, and the Delta region.

At 02:47:58 — 45 seconds after the first alert — the Amdocs billing engine (SV-003) encrypts its own data directory. A ransom note appears on the admin console:

This is not a drill. This is Operation Blackout — a coordinated, multi-vector attack combining ransomware encryption with BGP route injection to cause maximum chaos at the moment of minimum human capacity.

What happens next is a direct test of every document, every procedure, and every decision Samara Risha's team has prepared.

The Stakes

Financial Exposure	Regulatory Risk	Reputational Risk
EGP 470,000+/hour (Core Network + Mobile + Enterprise combined — per BIA 06-KNKT-BIA)	NTRA mandatory incident notification within 4 hours. Failure = license review + fines.	2.3 million subscribers without connectivity. National media coverage inevitable.

PHASE 1 — DETECTION & IDENTIFICATION ⏱ 0–30 MIN

Documents Activated

10-KNKT-BCP-CIR (Cyber IR) · 04-KNKT-ASSET (Asset Inventory) · 08-KNKT-CONT (Emergency Contacts) · 06-KNKT-BIA (BIA)

What Happens

The NOC Shift Lead — trained under TRN-004 (SOC & IR Training, doc 15-KNKT-TRAIN) — does not panic. He has run this tabletop twice (doc 12-KNKT-DRTEST). He knows the first step.

Time	Actor	Action	Document Used
02:48	NOC Shift Lead	QRadar SIEM confirms 47 true-positive alerts. BGP peer loss confirmed. Flags as potential P1.	10-KNKT-BCP-CIR §3.1 Step 1-2 / SV-005 (Asset Inv.)
02:50	SOC Analyst	Identifies ransomware IOCs: encrypted file extensions on SV-003. Lateral movement detected toward SV-001.	10-KNKT-BCP-CIR §3.1 Step 2
02:53	SOC Lead	Classifies incident: P1 Critical. Dual incident — cyber + network. Calls CISO Samara Risha on mobile.	10-KNKT-BCP-CIR §3.1 Step 3 / 08-KNKT-CONT T1
03:02	CISO (Samara Risha)	Confirms P1 classification. Opens INC-2026-001 in ServiceNow. Activates Crisis Management Team.	10-KNKT-BCP-CIR §3.1 Step 4
03:05	CISO	References BIA (06-KNKT-BIA) BF-001 & BF-002. MTD = 4hrs. RTO target = 1hr. Clock is running.	06-KNKT-BIA BF-001, BF-002, BF-007

The Asset Inventory Saves 20 Minutes

Because Asset Inventory (04-KNKT-ASSET) is pre-built and maintained, the SOC Lead immediately knows: which systems are affected (NI-001, SV-003, SV-001), who owns them (NOC / IT Director / Finance), and which backup systems exist (NI-004 DR Router, SV-007 DR Compute). Without this document, the team would spend 20+ minutes just answering: 'what do we actually have?'

The BIA Gives the Team Their Priorities

Samara pulls up BIA document 06-KNKT-BIA. In 90 seconds she has the answer that matters most under pressure:

- BF-001 Core Network: RTO ≤ 1 hour. Financial impact EGP 200K+/hr. TRA mandatory reporting. PRIORITY 1.
- BF-002 Mobile Services: RTO ≤ 2 hours. EGP 150K+/hr. PRIORITY 1.
- BF-004 Billing: RTO ≤ 4 hours. EGP 80K+/hr. Can wait — but not long. PRIORITY 2.
- BF-010 HR & Payroll: RTO ≤ 24 hours. Handle last.

This is exactly what a BIA is for. Not a theoretical exercise — a real-time decision support tool.

PHASE 2 — CONTAINMENT & NOTIFICATION ⏱ 30 MIN – 2 HRS

Documents Activated

10-KNKT-BCP-CIR §3.2 · 09-KNKT-BCP-NET §4.1–4.2 · 07-KNKT-CCP · 08-KNKT-CONT · 02-KNKT-STKS

Containment — Stopping the Bleeding

Two parallel tracks run simultaneously. Track A is the Cyber Incident Response procedure (10-KNKT-BCP-CIR). Track B is the Network BCP (09-KNKT-BCP-NET). Both are running at the same time, led by two separate team leads reporting to Samara.

TRACK A — Cyber IR (10-KNKT-BCP-CIR §3.2)	TRACK B — Network BCP (09-KNKT-BCP-NET §4.2)
03:17 · SOC Lead isolates SV-003 and SV-001 at Palo Alto NGFW (NI-005). Revokes all service account credentials in Active Directory. 03:32 · Sys Admin verifies Veeam immutable backups pre-dating infection. Backups confirmed clean. Evidence preservation begins — memory dumps taken. NO POWER OFF rule enforced. 03:45 · CISO invokes NO RANSOM PAYMENT policy. External CIRT retainer (T4 contact, doc 08-KNKT-CONT) engaged for forensic support.	03:15 · Network Engineer SSHs to DR Failover Router NI-004 (6th October DR site). Activates pre-configured DR BGP profile. 03:28 · BGP sessions re-established with two upstream ISP peers via DR path. Partial subscriber traffic rerouted. 03:47 · NMS (AP-002) confirms 68% subscriber traffic restored via DR site. Third BGP peer pending. CISO notified.

Notification — The Crisis Communication Machine

While containment is underway, the Crisis Communication Plan (07-KNKT-CCP) activates in parallel. This is not optional — it is a choreographed, time-bound sequence defined before the incident ever happened.

Time	Stakeholder	Channel	Owner	Doc Reference
03:02	Board of Directors	Encrypted WhatsApp (CEO-controlled group)	CEO	07-KNKT-CCP §4 / 02-KNKT-STKS
03:10	IT Director, NOC, SOC	Internal Slack + phone bridge	CISO	07-KNKT-CCP §4 / 08-KNKT-CONT T2
03:45	TRA (Regulator)	Secure email + phone (Regulatory Notification Template)	Compliance Manager	07-KNKT-CCP §6.2 — within 4hr mandate
04:10	Enterprise Customers	Account manager direct call + email	Customer Care Director	07-KNKT-CCP §4 — SLA obligation ≤2hr
04:30	2.3M Subscribers	SMS broadcast + IVR + status.konekttelcom.com	Customer Care Director	07-KNKT-CCP §5 — within 4hr target
04:47	TRA — Formal Notice Filed	Written incident report with timestamp log	Compliance + Legal	08-KNKT-CONT T4 — logged in ServiceNow

The 4-Hour TRA Window — Hit with 13 Minutes to Spare

TRA mandatory incident notification must be filed within 4 hours of detection. Detection: 02:47. TRA notification filed: 04:47. Time taken: 4 hours flat — compliant. The Crisis Communication Plan (07-KNKT-CCP) and Emergency Contact Tree (08-KNKT-CONT) made this possible by defining the exact path before the fire started.

PHASE 3 — DISASTER RECOVERY ⏱ 2–8 HRS

Documents Activated

09-KNKT-BCP-NET §4.3–4.4 · 10-KNKT-BCP-CIR §3.3 · 11-KNKT-BCP-BILL · 06-KNKT-BIA (Recovery Priorities)

DR Site Full Activation — 6th October City

At 04:00, with 43 minutes elapsed since containment began, the NOC Manager makes the call: Phase 4 DR Site Full Failover (09-KNKT-BCP-NET §4.4). The Primary Data Centre in Nasr City is compromised. The 6th October DR site becomes the operational centre of Konekt Telcom.

- VMware vSphere DR Compute Cluster (SV-007) brought fully online — all critical VMs spun up
- Core IP/MPLS routing fully migrated to DR site — all three BGP peers re-established by 04:38
- ISMS Management Server (SV-005 / QRadar SIEM) restored from clean backup — SOC visibility restored
- Subscriber traffic: 100% rerouted via DR MPLS fabric by 05:12 — RTO for BF-001: 2 hours 25 minutes

RTO RESULT — Core Network (BF-001)

Target RTO: ≤ 1 hour (escalation threshold: 45 min → CEO notified ✓ done at 03:17). Actual RTO achieved: 2 hours 25 minutes. MTD = 4 hours. RTO missed but MTD not breached — the escalation procedures in 09-KNKT-BCP-NET triggered CEO notification exactly as designed. No regulatory breach.

Billing Recovery — BCP Document 11 Saves Revenue

While network recovery dominates, the Billing team works quietly but critically in the background. The Amdocs billing engine (SV-003) was the first system encrypted. A CDR data gap of 1 hour 47 minutes exists — every unbilled call, SMS, and data session in that window.

- 11-KNKT-BCP-BILL §4.2: Amdocs hot standby server activated. Billing engine operational by 06:15.
- CDR mediation layer queued records recovered from immutable backup. Gap: 1hr 47min — triggering financial audit per 11-KNKT-BCP-BILL §3 (RPO >1hr threshold).
- Finance Director notified. Manual billing hold placed on affected subscriber accounts pending reconciliation.
- Actual Billing RTO: 3 hours 28 minutes. Target: ≤ 4 hours. ☑ TARGET MET.

Eradication — Removing the Threat

Parallel to DR operations, the SOC team and external CIRT execute the eradication phase (10-KNKT-BCP-CIR §3.3). This is the phase that cannot be rushed — every step missed here means the attacker stays in the environment.

- All affected systems (SV-003, SV-001 partial) re-imaged from clean baseline images.
- All credentials reset across the environment — 847 accounts, 23 service accounts, 6 API keys.

- Full threat hunt conducted across all systems for persistence mechanisms and lateral movement.
- IOC blocks pushed to Palo Alto NGFW (NI-005): 34 malicious IPs, 8 domains, 3 file hashes.
- EG-CERT (Egypt CERT) notified per 08-KNKT-CONT T4. Intelligence shared.

PHASE 4 — RECOVERY & RESTORATION ⏱ 8–24 HRS

Documents Activated
06-KNKT-BIA (Priority Order) · 10-KNKT-BCP-CIR §3.4 · 07-KNKT-CCP §6.3 (Resolution Notice)

Restoring Services in Priority Order

The BIA (06-KNKT-BIA §4) mandates the exact order of recovery. This is not guesswork. Every hour spent restoring a P3 system while a P1 system waits costs EGP 200,000. Samara enforces the priority sequence:

Priority	Business Function	BIA Ref.	RTO Target	RTO Actual	Status	RPO Met?
P1	Core Network Operations	BF-001	≤ 1 hr	2h 25m	MTD Met ✓	Yes ✓
P1	Mobile Services (Voice & Data)	BF-002	≤ 2 hr	3h 10m	MTD Met ✓	Yes ✓
P1	Enterprise Connectivity (MPLS/SD-WAN)	BF-007	≤ 2 hr	4h 05m	MTD Met ✓	Yes ✓
P2	Billing & Revenue Assurance	BF-004	≤ 4 hr	3h 28m ✓	RTO Met ✓	Audit req.
P2	Security Operations (SOC/SIEM)	BF-008	≤ 2 hr	1h 52m ✓	RTO Met ✓	Yes ✓
P3	Customer Self-Service Portal	BF-006	≤ 8 hr	7h 15m ✓	RTO Met ✓	Yes ✓
P4	HR & Payroll	BF-010	≤ 24 hr	18h 00m ✓	RTO Met ✓	Yes ✓

Resolution & Public Communication

At 19:30 on the same day — 16 hours and 43 minutes after the first alert — Samara Risha confirms full service restoration. The Resolution Notice template from Crisis Communication Plan (07-KNKT-CCP §6.3) is deployed:

Resolution Notice — Published 19:30 via status.konekttelcom.com + SMS broadcast
"Konekt Telcom is pleased to confirm that full service has been restored as of 19:30 on 16/04/2026. The disruption was caused by a sophisticated external cyberattack on our primary data centre. We have successfully restored all services from our disaster recovery infrastructure. We are conducting a full post-incident review and have implemented immediate protective measures. We sincerely apologise to our subscribers and enterprise customers for the disruption. Reference: INC-2026-001."

PHASE 5 — POST-INCIDENT REVIEW & IMPROVEMENT ⏱ DAY 2–10

Documents Activated
13-KNKT-PIR · 12-KNKT-DRTEST · 14-KNKT-SOA · 15-KNKT-TRAIN

Post-Incident Review — Document 13

Within 5 business days, CISO Samara Risha completes Post-Incident Review INC-2026-001 using template 13-KNKT-PIR. This is not a blame session — it is the program's learning engine.

PIR Section	Findings — INC-2026-001
Root Cause	Ransomware delivered via spear-phishing email to NOC engineer 6 days prior. Malware dormant, activated at 02:47. Email security control A.8.23 (14-KNKT-SOA) rated 'Partial' — gap confirmed.
What Worked	Emergency Contact Tree (08-KNKT-CONT) — all T1 contacts reached within 18 minutes. BIA (06-KNKT-BIA) — recovery priorities followed exactly. BCP-NET DR failover — executed as rehearsed. TRA notification — filed within regulatory window.
What Failed	Core Network RTO missed by 85 minutes (target 1hr, actual 2hr 25min). BGP DR profile had an outdated ISP peer IP — caused 40-minute delay. CDR billing RPO exceeded by 47 minutes — financial audit required.
Corrective Actions	CA-001: DR BGP profile peer IPs verified quarterly (IT Director, 30 days). CA-002: Email security uplift — implement Advanced Threat Protection (A.8.23 to 'Full', 14-KNKT-SOA). CA-003: Conduct full DR failover test within 90 days (12-KNKT-DRTEST). CA-004: TRN-004 IR training refresher for all SOC staff (15-KNKT-TRAIN).

DR Test Program— Document 12

The DR Test Plan (12-KNKT-DRTEST) now records Operation Blackout as an unplanned live test — the highest fidelity test possible. Results are formally logged. The following scheduled tests are brought forward:

- Full Interruption Test — core network — moved from Q4 to Q3 2026 (90-day corrective action)
- Parallel Billing Test — Amdocs hot standby validation — scheduled within 60 days
- Red Team Exercise — social engineering + phishing simulation for all staff — Q2 2026

Statement of Applicability — Document 14

The SoA (14-KNKT-SOA) is immediately updated. Control A.8.23 (Web Filtering / Email Security) status changes from 'Partial' to 'Planned — Remediation in Progress'. The incident is cited as the business justification. ISO 27001 auditors will see not a gap — but a functioning improvement cycle.

Training — Document 15 Closes the Human Gap

The root cause was human: a phishing email was clicked. The Training Plan (15-KNKT-TRAIN) is updated:

- TRN-002 (Phishing Awareness) — frequency increased from Annual to Bi-annual for all staff
- TRN-004 (SOC & IR Training) — mandatory refresher within 30 days for all SOC personnel

- TRN-005 (BCP Tabletop) — specific ransomware + BGP attack scenario added to next exercise

PROGRAMME ASSESSMENT — DID THE FRAMEWORK HOLD?

Assessment Criterion	Result	Evidence
P1 incident detected and classified within 30 min	✅ PASS	INC-2026-001: CISO notified at 03:02 — 15 min from first alert
TRA regulatory notification within 4 hours	✅ PASS	Filed 04:47 — 4 hours flat; logged in ServiceNow with timestamp
No ransom paid; NO PAYMENT policy enforced	✅ PASS	10-KNKT-BCP-CIR §4 — CISO decision logged at 03:45; CIRT engaged
Subscriber communications issued within 4 hours	✅ PASS	SMS broadcast + status page live by 04:30 — 1hr 43min within target
MTD not breached for any P1 function	✅ PASS	All P1 MTDs = 4hr. Full DR site operational by 05:12 — within MTD
Core Network RTO ≤ 1 hour met	⚠ MISS	Actual 2h 25min. BGP profile error caused delay. Corrective action CA-001 raised.
Billing RPO ≤ 1 hour met	⚠ MISS	CDR gap 1hr 47min. Financial audit initiated per 11-KNKT-BCP-BILL. Corrective action CA-003.
ISO 27001 PIR completed within 5 business days	✅ PASS	PIR filed Day 5. Board briefing Day 8. TRA closure report Day 10. All within mandate.
Zero data exfiltration confirmed	✅ PASS	CIRT forensic analysis confirmed encryption-only; no exfiltration channels detected in SIEM

FINAL VERDICT — The Program Worked

7 out of 9 criteria passed outright. 2 RTOs missed — but both were caught by escalation procedures built into the BCP documents themselves. No regulatory breach. No data loss. No ransom paid. 2.3 million subscribers reconnected in under 17 hours. The two misses have generated concrete corrective actions that will make the program stronger. This is not failure — this is the system working exactly as designed.

CONCLUSION

Operation Blackout was the worst-case scenario Konekt Telcom's BC/DR/IRprogram was built for. A coordinated, simultaneous ransomware attack and network outage — hitting at 02:47 on a Tuesday when the building is half-empty and the attackers expect chaos.

They got a playbook instead.

Every document in this 15-documentprogram contributed a specific, measurable value during the incident:

- The BIA told the team what to fix first — saving hours of misdirected effort
- The Asset Inventory told them what they were dealing with — in seconds, not hours
- The BCP documents gave step-by-step recovery actions — removing guesswork under pressure
- The Crisis Communication Plan ensured regulators, customers, and the board were never in the dark
- The Emergency Contact Tree meant every call was made in the right order to the right person
- The PIR template turned the crisis into institutional knowledge
- The Training Plan explained how the human failure happened — and how to close it

BC/DR/IR is not about preventing every incident. It is about ensuring that when the worst happens, your organisation has the clarity, the tools, and the trained people to respond with precision — and emerge with its customers' trust intact.

Konekt Telcom's Programme Status Post-Operation Blackout

Programme maturity: TESTED UNDER LIVE P1 CONDITIONS. Two corrective actions in progress. Next scheduled DR test brought forward to Q3 2026. ISO 27001 SoA updated. TRA notified and engaged. Board briefed. Subscribers reconnected. Lessons documented. Programme stronger.

VERSION HISTORY

Version	Date	Description	Author	Approved By
1.0	16/04/2026	Initial Use Case Document — Operation Blackout	Samara Risha	CISO / CEO